

KIRLOSKAR CHILLERS PRIVATE LIMITED

A Kirloskar Group Company

Document Name	KWS, R-134a HEAT PUMP RATING CHART				
Document No.	ESP/14-15/05	Revision	R02	Date	28-Aug-17
Prepared by	AHW	Approved by	SSN	No. of sheets	5

KWS Ratings, R-134a Application

Unit Model	Compressor Model	Leaving Chilled Water Temp °C	Leaving Condenser Water Temperature °C											
			45			50			55			60		
			Cooling Capacity in TR	Power Input in KW	Heating Capacity in TR	Cooling Capacity in TR	Power Input in KW	Heating Capacity in TR	Cooling Capacity in TR	Power Input in KW	Heating Capacity in TR	Cooling Capacity in TR	Power Input in KW	Heating Capacity in TR
KWS 050.14	CXH03 80 264Y	5.00	39.3	43.5	51.7	36.85	47.4	50.3	33.9	52.3	48.8	31.0	58.4	47.6
		6.00	40.6	43.9	53.0	37.99	47.9	51.6	34.9	52.8	50.0	32.0	59.0	48.8
		7.00	42.3	44.4	54.9	39.6	48.4	53.4	36.5	53.3	51.6	33.4	59.6	50.3
		8.00	43.6	44.8	56.3	40.8	48.8	54.7	37.5	53.8	52.8	34.4	60.1	51.5
		9.00	45.3	45.3	58.1	42.4	49.3	56.4	39.0	54.4	54.5	35.7	60.8	53.0
		10.00	47.0	45.7	59.9	44.0	49.8	58.1	40.5	54.9	56.1	37.0	61.4	54.5
KWS 060.14	CXH01 80 298Y	5.00	44.4	49.1	58.4	41.60	53.5	56.8	38.3	59.0	55.0	35.0	66.0	53.8
		6.00	45.8	49.6	59.9	42.88	54.0	58.2	39.4	59.6	56.4	36.1	66.6	55.0
		7.00	47.7	50.1	62.0	44.7	54.6	60.2	41.1	60.2	58.3	37.6	67.3	56.8
		8.00	49.2	50.5	63.6	46.1	55.1	61.7	42.4	60.7	59.7	38.8	67.9	58.1
		9.00	51.1	51.1	65.6	47.9	55.7	63.7	44.0	61.4	61.5	40.3	68.7	59.8
		10.00	53.0	51.6	67.7	49.6	56.2	65.6	45.7	62.0	63.3	41.8	69.3	61.5
KWS 080.14	CXH51 110 398Y	5.00	63.3	67.4	82.5	59.29	73.4	80.2	54.5	81.0	77.6	49.9	90.6	75.7
		6.00	65.2	68.1	84.6	61.12	74.2	82.2	56.2	81.8	79.5	51.4	91.5	77.5
		7.00	68.1	68.8	87.6	63.8	74.9	85.1	58.6	82.6	82.2	53.6	92.5	79.9
		8.00	70.1	69.4	89.8	65.7	75.6	87.2	60.4	83.4	84.1	55.2	93.3	81.8
		9.00	72.8	70.1	92.8	68.2	76.4	90.0	62.8	84.3	86.7	57.4	94.3	84.2
		10.00	75.5	70.8	95.7	70.8	77.2	92.7	65.1	85.1	89.3	59.5	95.3	86.6
KWS 090.14	CXH53 110 468Y	5.00	72.1	76.4	93.8	67.54	83.3	91.2	62.1	91.8	88.3	56.8	102.8	86.1
		6.00	74.3	77.2	96.3	69.63	84.1	93.6	64.1	92.8	90.4	58.6	103.8	88.1
		7.00	77.5	78.0	99.7	72.6	85.0	96.8	66.8	93.7	93.5	61.1	104.9	90.9
		8.00	79.9	78.7	102.2	74.8	85.7	99.2	68.8	94.6	95.7	62.9	105.8	93.0
		9.00	83.0	79.5	105.6	77.7	86.7	102.4	71.5	95.6	98.7	65.4	107.0	95.8
		10.00	86.1	80.3	108.9	80.6	87.5	105.5	74.2	96.5	101.6	67.8	108.0	98.6

- Notes:
- 1) Cooling capacity and power input are based on $\Delta T = 5^{\circ}\text{C}$ for entering/leaving chilled water temperature and $\Delta T = 5^{\circ}\text{C}$ for entering/leaving condenser water temperature
 - 2) Total suction superheat is 5°C and subcooling is 5°C
 - 3) Evaporator fouling factor = $0.0001 \text{ ft}^2\text{hr}^{\circ}\text{F}/\text{BTU}$; Condenser fouling factor = $0.00025 \text{ ft}^2\text{hr}^{\circ}\text{F}/\text{BTU}$

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			Cooling Capacity in TR	Power Input in KW	Heating Capacity in TR	Cooling Capacity in TR	Power Input in KW	Heating Capacity in TR	Cooling Capacity in TR	Power Input in KW	Heating Capacity in TR	Cooling Capacity in TR	Power Input in KW	Heating Capacity in TR
KWS 110.14	CXH53 110 538Y	5.00	83.2	89.0	108.6	77.98	97.0	105.6	71.7	107.0	102.2	65.7	119.7	99.7
		6.00	85.8	89.9	111.4	80.39	98.0	108.3	74.0	108.0	104.7	67.7	120.8	102.1
		7.00	89.5	90.8	115.3	83.9	99.0	112.0	77.1	109.2	108.2	70.6	122.1	105.3
		8.00	92.2	91.6	118.3	86.4	99.9	114.8	79.5	110.1	110.8	72.7	123.2	107.8
		9.00	95.8	92.6	122.1	89.7	101.0	118.4	82.5	111.3	114.2	75.5	124.5	111.0
		10.00	99.4	93.5	126.0	93.1	101.9	122.1	85.6	112.4	117.6	78.4	125.8	114.1
KWS 130.14	CXH91 160 620Y	5.00	95.9	101.5	124.8	89.86	110.6	121.3	82.7	122.0	117.4	75.6	136.5	114.4
		6.00	98.9	102.5	128.0	92.63	111.7	124.4	85.2	123.2	120.3	77.9	137.9	117.2
		7.00	103.1	103.6	132.6	96.6	112.9	128.7	88.9	124.5	124.3	81.3	139.3	120.9
		8.00	106.2	104.5	136.0	99.5	113.9	131.9	91.6	125.6	127.3	83.7	140.6	123.7
		9.00	110.4	105.7	140.4	103.4	115.2	136.1	95.1	127.0	131.2	87.0	142.1	127.4
		10.00	114.5	106.7	144.8	107.2	116.3	140.3	98.7	128.2	135.1	90.2	143.5	131.0
KWS 165.14	CXH91 210 810Y	5.00	126.3	134.4	164.5	118.30	146.5	160.0	108.8	161.5	154.8	99.5	180.8	150.9
		6.00	130.2	135.8	168.8	121.95	148.0	164.0	112.2	163.2	158.6	102.6	182.6	154.5
		7.00	135.8	137.2	174.8	127.2	149.5	169.7	117.0	164.8	163.9	107.0	184.4	159.5
		8.00	139.9	138.4	179.2	131.0	150.8	173.9	120.5	166.3	167.8	110.2	186.1	163.2
		9.00	145.3	139.9	185.1	136.1	152.5	179.5	125.2	168.1	173.0	114.5	188.1	168.0
		10.00	150.7	141.3	190.9	141.2	154.0	185.0	129.9	169.8	178.2	118.8	190.0	172.8
KWS 180.14	CXH93 210 912Y	5.00	138.4	147.2	180.3	129.64	160.5	175.3	119.3	177.0	169.6	109.1	198.0	165.4
		6.00	142.7	148.7	185.0	133.64	162.1	179.7	122.9	178.7	173.8	112.5	200.0	169.4
		7.00	148.8	150.3	191.5	139.4	163.7	186.0	128.2	180.6	179.6	117.3	202.1	174.8
		8.00	153.3	151.6	196.4	143.6	165.2	190.6	132.1	182.2	183.9	120.8	203.9	178.8
		9.00	159.2	153.3	202.8	149.2	167.0	196.7	137.2	184.2	189.6	125.5	206.1	184.1
		10.00	165.2	154.8	209.2	154.7	168.7	202.7	142.3	186.0	195.2	130.2	208.1	189.4

- Notes:
- 1) Cooling capacity and power input are based on $\Delta T = 5^{\circ}\text{C}$ for entering/leaving chilled water temperature and $\Delta T = 5^{\circ}\text{C}$ for entering/leaving condenser water temperature
 - 2) Total suction superheat is 5°C and subcooling is 5°C
 - 3) Evaporator fouling factor = $0.0001 \text{ ft}^2\text{hr}^{\circ}\text{F}/\text{BTU}$; Condenser fouling factor = $0.00025 \text{ ft}^2\text{hr}^{\circ}\text{F}/\text{BTU}$

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KWS Ratings, R-134a Application

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			45			50			55			60		
			Cooling Capacity in TR	Power Input in KW	Heating Capacity in TR	Cooling Capacity in TR	Power Input in KW	Heating Capacity in TR	Cooling Capacity in TR	Power Input in KW	Heating Capacity in TR	Cooling Capacity in TR	Power Input in KW	Heating Capacity in TR
KWS 100.24	CXH03 80 264Y X 2	5.00	78.7	86.9	103.4	73.76	94.7	100.7	67.9	104.4	97.5	62.1	116.8	95.3
		6.00	81.2	87.8	106.1	76.03	95.6	103.2	69.9	105.5	99.9	64.0	118.0	97.5
		7.00	84.7	88.7	109.9	79.3	96.6	106.8	73.0	106.6	103.3	66.7	119.2	100.6
		8.00	87.2	89.5	112.6	81.7	97.5	109.4	75.1	107.5	105.7	68.7	120.3	103.0
		9.00	90.6	90.4	116.3	84.9	98.6	112.9	78.1	108.7	109.0	71.4	121.6	106.0
		10.00	94.0	91.3	119.9	88.0	99.5	116.3	81.0	109.8	112.2	74.1	122.8	109.0
KWS 130.24	CXH03 80 340Y X 2	5.00	104.3	111.8	136.1	97.75	121.8	132.4	89.9	134.4	128.1	82.3	150.3	125.0
		6.00	107.6	112.9	139.7	100.76	123.1	135.8	92.7	135.7	131.3	84.8	151.8	128.0
		7.00	112.2	114.1	144.6	105.1	124.3	140.5	96.7	137.1	135.7	88.5	153.4	132.1
		8.00	115.6	115.1	148.3	108.3	125.5	143.9	99.6	138.4	138.9	91.1	154.8	135.1
		9.00	120.1	116.4	153.2	112.5	126.8	148.5	103.5	139.9	143.2	94.6	156.5	139.1
		10.00	124.5	117.5	158.0	116.7	128.1	153.1	107.3	141.2	147.5	98.2	158.0	143.1
KWS 145.24	CXH51 110 398Y + CXH03 80 340Y	5.00	115.4	123.3	150.5	108.14	134.4	146.4	99.5	148.2	141.6	90.5	165.8	137.6
		6.00	119.0	124.5	154.4	111.48	135.7	150.1	102.6	149.7	145.1	93.3	167.5	140.9
		7.00	124.1	125.8	159.9	116.3	137.1	155.3	107.0	151.2	150.0	97.3	169.2	145.4
		8.00	127.9	127.0	164.0	119.8	138.4	159.1	110.2	152.6	153.6	100.2	170.7	148.8
		9.00	132.8	128.3	169.3	124.4	139.9	164.2	114.5	154.2	158.3	104.1	172.6	153.2
		10.00	137.8	129.6	174.6	129.1	141.2	169.2	118.7	155.8	163.0	108.0	174.3	157.6
KWS 160.24	CXH51 110 398Y X 2	5.00	126.5	134.8	164.9	118.54	146.9	160.3	109.1	162.0	155.1	99.8	181.3	151.3
		6.00	130.4	136.1	169.2	122.20	148.4	164.4	112.4	163.6	159.0	102.8	183.1	154.9
		7.00	136.1	137.5	175.2	127.5	149.9	170.1	117.3	165.3	164.3	107.3	185.0	159.9
		8.00	140.2	138.8	179.6	131.3	151.2	174.3	120.8	166.8	168.2	110.5	186.6	163.6
		9.00	145.6	140.3	185.5	136.4	152.9	179.9	125.5	168.6	173.4	114.8	188.7	168.4
		10.00	151.0	141.7	191.3	141.5	154.4	185.4	130.2	170.3	178.6	119.1	190.5	173.3

- Notes:
- 1) Cooling capacity and power input are based on $\Delta T = 5^{\circ}\text{C}$ for entering/leaving chilled water temperature and $\Delta T = 5^{\circ}\text{C}$ for entering/leaving condenser water temperature
 - 2) Total suction superheat is 5°C and subcooling is 5°C
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			45			50			55			60		
			Cooling Capacity in TR	Power Input in KW	Heating Capacity in TR	Cooling Capacity in TR	Power Input in KW	Heating Capacity in TR	Cooling Capacity in TR	Power Input in KW	Heating Capacity in TR	Cooling Capacity in TR	Power Input in KW	Heating Capacity in TR
KWS 185.24	CXH53 110 468Y X 2	5.00	144.2	152.9	187.7	135.07	166.6	182.4	124.3	183.7	176.5	113.7	205.6	172.1
		6.00	148.6	154.4	192.5	139.24	168.2	187.1	128.1	185.5	180.9	117.2	207.6	176.2
		7.00	155.0	156.0	199.4	145.2	170.0	193.6	133.6	187.5	186.9	122.2	209.7	181.9
		8.00	159.7	157.4	204.5	149.6	171.5	198.4	137.6	189.1	191.4	125.9	211.6	186.1
		9.00	165.9	159.1	211.1	155.4	173.4	204.7	143.0	191.2	197.3	130.8	213.9	191.6
		10.00	172.1	160.7	217.8	161.2	175.1	211.0	148.3	193.1	203.2	135.7	216.0	197.1
KWS 215.24	CXH53 110 538Y X 2	5.00	166.6	178.0	217.2	156.04	193.9	211.2	143.5	213.9	204.4	131.3	239.3	199.4
		6.00	171.7	179.7	222.8	160.86	195.9	216.6	148.0	216.0	209.4	135.4	241.7	204.1
		7.00	179.1	181.6	230.8	167.8	197.9	224.1	154.4	218.2	216.4	141.2	244.2	210.6
		8.00	184.5	183.2	236.6	172.8	199.7	229.6	159.0	220.2	221.6	145.4	246.4	215.5
		9.00	191.6	185.2	244.3	179.5	201.8	236.9	165.2	222.6	228.5	151.1	249.1	221.9
		10.00	198.8	187.0	252.0	186.2	203.8	244.2	171.3	224.8	235.3	156.7	251.5	228.3
KWS 235.24	CXH53 110 538Y + CXH91 160 620Y	5.00	179.2	190.5	233.4	167.86	207.6	226.9	154.4	229.0	219.5	141.3	256.2	214.1
		6.00	184.7	192.4	239.4	173.04	209.7	232.7	159.2	231.2	225.0	145.6	258.7	219.2
		7.00	192.7	194.4	248.0	180.5	211.8	240.7	166.0	233.6	232.5	151.9	261.4	226.2
		8.00	198.5	196.1	254.2	185.9	213.7	246.7	171.0	235.7	238.1	156.5	263.8	231.5
		9.00	206.2	198.3	262.6	193.1	216.1	254.6	177.7	238.3	245.4	162.5	266.6	238.4
		10.00	213.9	200.2	270.8	200.3	218.2	262.4	184.3	240.6	252.7	168.6	269.2	245.2
KWS 275.24	CXH91 160 620Y + CXH93 160 702Y	5.00	207.8	220.2	270.4	194.68	239.9	262.9	179.1	264.6	254.3	163.8	296.1	248.0
		6.00	214.2	222.4	277.5	200.69	242.3	269.6	184.6	267.2	260.6	168.9	299.0	253.9
		7.00	223.5	224.7	287.4	209.3	244.8	279.0	192.6	270.0	269.4	176.2	302.1	262.1
		8.00	230.2	226.7	294.6	215.6	247.0	285.9	198.3	272.4	275.8	181.5	304.8	268.2
		9.00	239.1	229.1	304.3	224.0	249.7	295.0	206.1	275.4	284.4	188.5	308.1	276.2
		10.00	248.0	231.4	313.9	232.4	252.2	304.1	213.8	278.1	292.8	195.6	311.2	284.1

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KWS 295.24	CXH93 160 702Y X 2	5.00	223.8	237.3	291.3	209.68	258.6	283.2	192.9	285.2	274.0	176.5	319.1	267.2
		6.00	230.7	239.6	298.9	216.15	261.2	290.4	198.8	288.0	280.8	181.9	322.3	273.6
		7.00	240.7	242.1	309.5	225.5	263.9	300.5	207.4	291.0	290.2	189.8	325.6	282.4
		8.00	247.9	244.3	317.4	232.2	266.2	307.9	213.6	293.6	297.1	195.5	328.5	288.9
		9.00	257.5	247.0	327.8	241.2	269.1	317.8	221.9	296.8	306.3	203.1	332.1	297.5
		10.00	267.2	249.4	338.1	250.3	271.8	327.6	230.2	299.7	315.5	210.6	335.3	306.0
KWS 330.24	CXH91 210 810Y X 2	5.00	252.5	268.8	329.0	236.55	293.0	319.9	217.6	323.1	309.5	199.1	361.5	301.9
		6.00	260.3	271.5	337.5	243.85	295.9	328.0	224.3	326.3	317.1	205.2	365.1	309.1
		7.00	271.5	274.3	349.5	254.4	298.9	339.4	234.0	329.7	327.8	214.1	368.9	319.0
		8.00	279.7	276.8	358.4	262.0	301.6	347.8	241.0	332.7	335.6	220.5	372.2	326.3
		9.00	290.5	279.8	370.1	272.2	304.9	358.9	250.4	336.3	346.0	229.0	376.3	336.0
		10.00	301.4	282.5	381.8	282.3	307.9	369.9	259.7	339.6	356.3	237.6	380.0	345.7
KWS 360.24	CXH93 210 912Y X 2	5.00	276.8	294.5	360.6	259.30	320.9	350.6	238.5	353.9	339.2	218.2	396.0	330.8
		6.00	285.3	297.4	369.9	267.30	324.2	359.5	245.9	357.5	347.6	224.9	400.0	338.7
		7.00	297.6	300.5	383.1	278.8	327.5	372.0	256.5	361.2	359.2	234.6	404.1	349.6
		8.00	306.6	303.2	392.8	287.2	330.4	381.2	264.2	364.4	367.8	241.7	407.7	357.6
		9.00	318.5	306.5	405.6	298.3	334.0	393.3	274.4	368.4	379.2	251.1	412.2	368.3
		10.00	330.4	309.5	418.4	309.5	337.3	405.4	284.7	372.0	390.5	260.4	416.2	378.8

- Notes:
- 1) Cooling capacity and power input are based on $\Delta T = 5^{\circ}\text{C}$ for entering/leaving chilled water temperature and $\Delta T = 5^{\circ}\text{C}$ for entering/leaving condenser water temperature
 - 2) Total suction superheat is 5°C and subcooling is 5°C
 - 3) Evaporator fouling factor = $0.0001 \text{ ft}^2\text{hr}^{\circ}\text{F}/\text{BTU}$; Condenser fouling factor = $0.00025 \text{ ft}^2\text{hr}^{\circ}\text{F}/\text{BTU}$